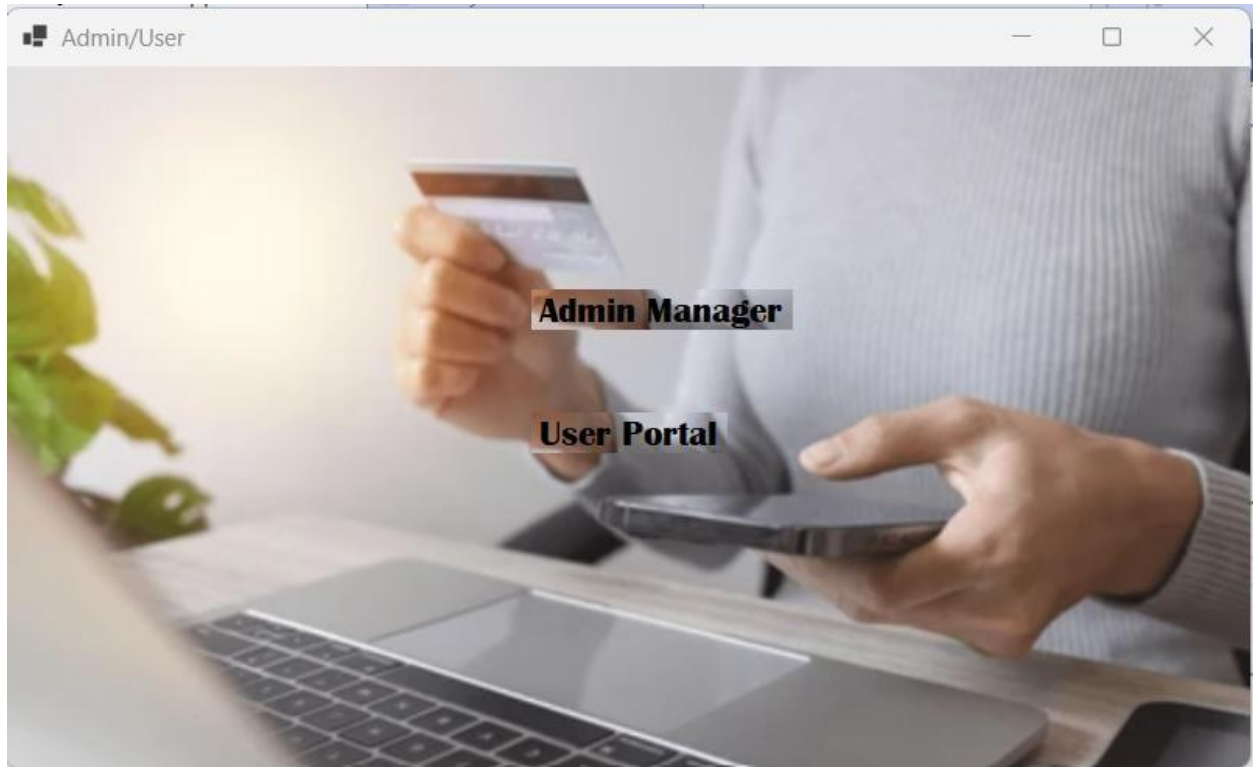


DB FINAL PROJECT

Front Interface:



Code:

```
Dfeag  
using System;  
using System.Collections.Generic;  
using System.ComponentModel;  
using System.Data;
```

```
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace WinFormsApp2
{
    public partial class Form7 : Form
    {
        public Form7()
        {
            InitializeComponent();
        }

        private void label2_Click(object sender, EventArgs e)
        {
            this.Hide();

            Form1 newForm = new Form1();

            // Show Form2
            newForm.Show();
        }

        private void label1_Click(object sender, EventArgs e)
        {
            this.Hide();
        }
    }
}
```

```
Form8 newForm = new Form8();

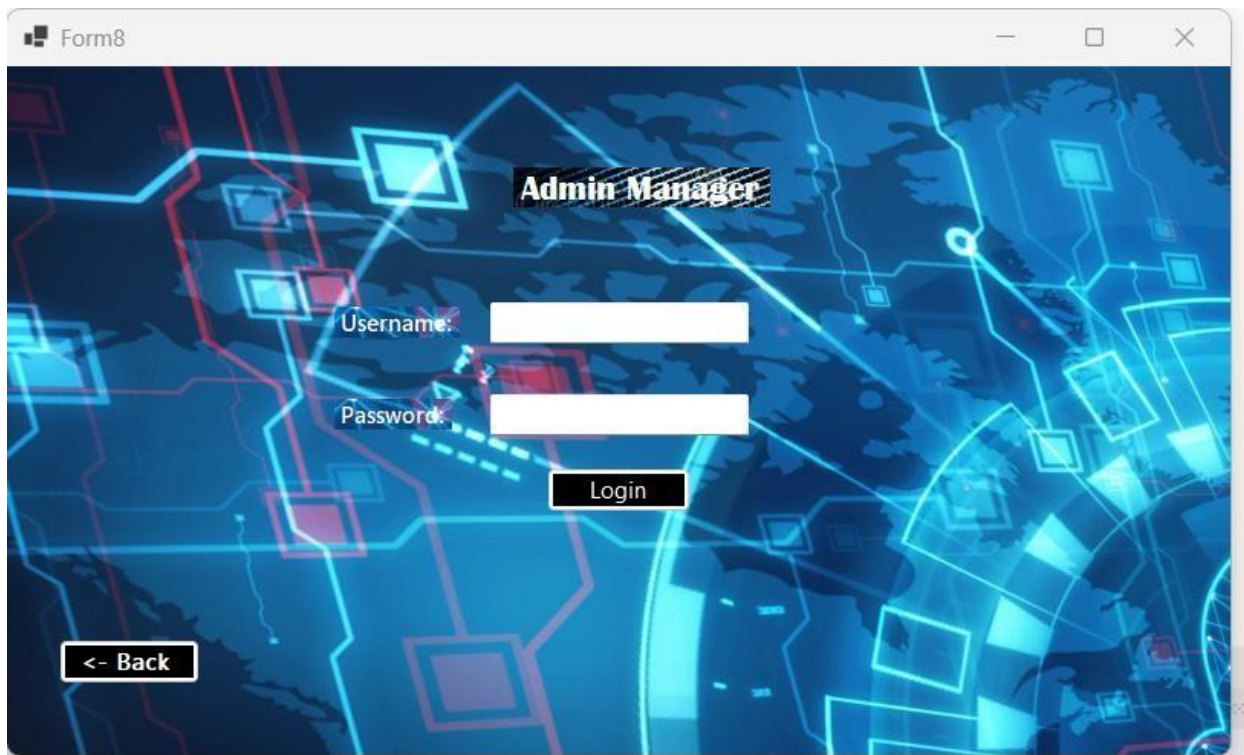
// Show Form2
newForm.Show();
}

private void Form7_Load(object sender, EventArgs e)
{

}

}
}
```

Admin Manager Interface:



Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace WinFormsApp2
{
    public partial class Form8 : Form
    {

```

```
public Form8()
{
    InitializeComponent();
}

private void button3_Click(object sender, EventArgs e)
{
    this.Hide();
    Form7 newForm = new Form7();

    // Show Form2
    newForm.Show();
}

private void button1_Click(object sender, EventArgs e)
{
    string username = textBox1.Text;
    string password = textBox2.Text;

    // Check if fields are empty
    if (string.IsNullOrEmpty(username) || string.IsNullOrEmpty(password))
    {
        MessageBox.Show("Please enter both username and password.", "Validation Error",
        MessageBoxButtons.OK, MessageBoxIcon.Warning);
        return;
    }
}
```

```
// Simulate authentication (Replace this with actual database verification)
if (username == "admin" && password == "12345")
{
    MessageBox.Show("Login Successful!", "Success", MessageBoxButtons.OK,
    MessageBoxIcon.Information);

    // Optionally, open a new form after successful login
    Form9 form2 = new Form9();
    form2.Show();
    this.Hide(); // Hides the current form
}
else
{
    MessageBox.Show("Invalid username or password. Please try again.", "Login Failed",
    MessageBoxButtons.OK, MessageBoxIcon.Error);
}
}

private void textBox1_TextChanged(object sender, EventArgs e)
{
    //username
    if (textBox1.Text.Length > 20)
    {
        MessageBox.Show("Username cannot exceed 20 characters.", "Input Error",
        MessageBoxButtons.OK, MessageBoxIcon.Warning);
        textBox1.Text = textBox1.Text.Substring(0, 20); // Trim excess characters
    }
}
```

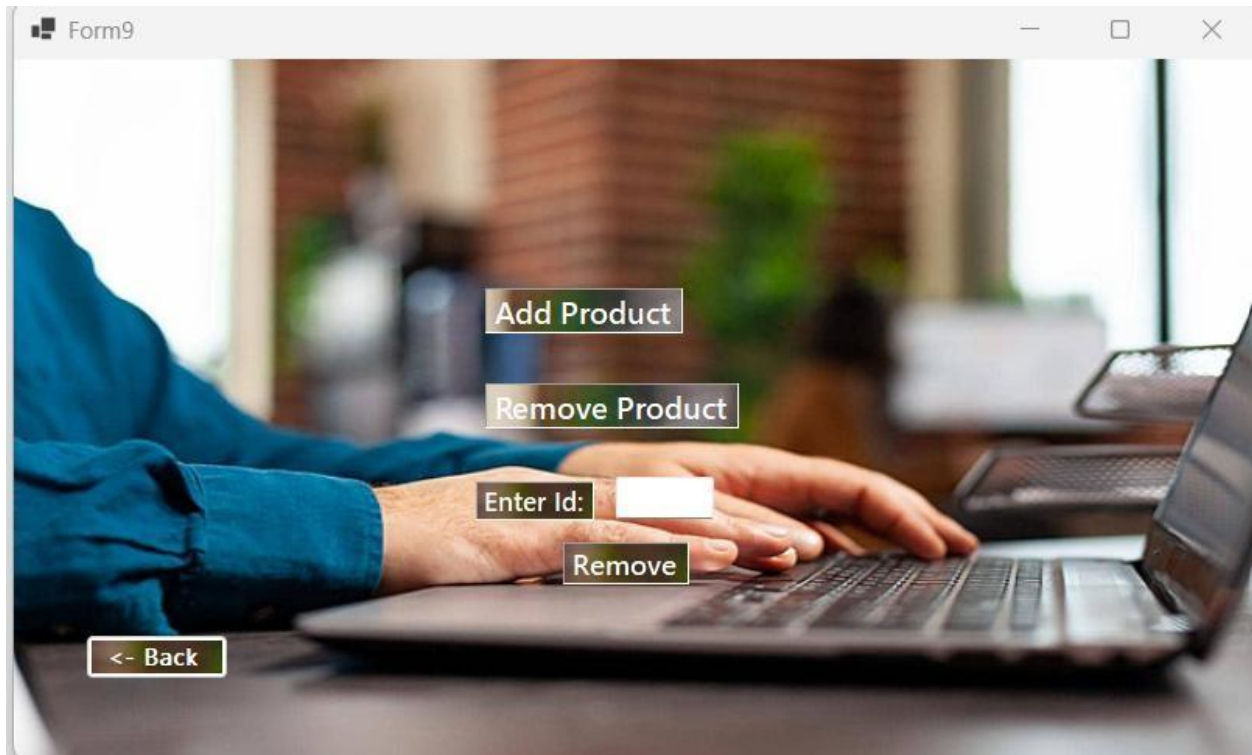
```
private void textBox2_TextChanged(object sender, EventArgs e)
{
    //password
    textBox2.PasswordChar = '*';
}

private void checkBox1_CheckedChanged(object sender, EventArgs e)
{
}

private void Form8_Load(object sender, EventArgs e)
{
}

}
}
```

Add product and remove product :



Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using MySql.Data.MySqlClient;

namespace WinFormsApp2
{
    public partial class Form9 : Form
```

```

{
    public Form9()
    {
        InitializeComponent();
    }

    private void label5_Click(object sender, EventArgs e)
    {
        // Get Product ID from textBox1
        int productId;
        if (!int.TryParse(textBox1.Text.Trim(), out productId) || productId <= 0)
        {
            MessageBox.Show("Please enter a valid Product ID.");
            return;
        }

        // Connection string
        string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
        MySqlConnection conn = new MySqlConnection(connstring);

        try
        {
            conn.Open();

            // Start a transaction to ensure both deletions are handled together
            MySqlTransaction transaction = conn.BeginTransaction();

            try

```

```

{
    // Delete the product from the `cart` table
    string deleteFromCartQuery = "DELETE FROM cart WHERE ProductID = @ProductID";
    MySqlCommand deleteFromCartCmd = new MySqlCommand(deleteFromCartQuery,
conn, transaction);
    deleteFromCartCmd.Parameters.AddWithValue("@ProductID", productId);
    deleteFromCartCmd.ExecuteNonQuery();

    // Delete the product from the `product` table
    string deleteFromProductQuery = "DELETE FROM product WHERE ProductID =
@ProductID";
    MySqlCommand deleteFromProductCmd = new
MySqlCommand(deleteFromProductQuery, conn, transaction);
    deleteFromProductCmd.Parameters.AddWithValue("@ProductID", productId);

    int rowsAffected = deleteFromProductCmd.ExecuteNonQuery();

    if (rowsAffected > 0)
    {
        transaction.Commit(); // Commit the transaction
        MessageBox.Show("Product and its associated cart entries deleted successfully.");

        // Clear the input field after deletion
        textBox1.Clear();
    }
    else
    {
        transaction.Rollback(); // Rollback the transaction
    }
}

```

```
        MessageBox.Show("No product found with the given Product ID.");
    }
}
catch
{
    transaction.Rollback(); // Rollback in case of an error
    throw;
}
}
catch (Exception ex)
{
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    conn.Close();
}
}

private void label1_Click(object sender, EventArgs e)
{
    this.Hide();
    Form10 newForm = new Form10();

    // Show Form2
    newForm.Show();
}
```

```
private void button3_Click(object sender, EventArgs e)
{
    this.Hide();
    Form8 newForm = new Form8();

    newForm.Show();
}

private void textBox1_TextChanged(object sender, EventArgs e)
{
}

private void Form9_Load(object sender, EventArgs e)
{
}

}
}
```

Add product :

The image shows a screenshot of a Windows application window titled "Form10". The window has a dark-themed background with a subtle abstract pattern. It contains four text input fields arranged vertically, each preceded by a label in a light green box: "Name:", "Price:", "Stock:", and "Description:". At the bottom of the form, there are two buttons: a button on the left labeled "<- Back" and a button on the right labeled "Add Product".

Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using MySql.Data.MySqlClient;

namespace WinFormsApp2
{
    public partial class Form10 : Form
```

```
{  
    public Form10()  
    {  
        InitializeComponent();  
    }  
  
    private void button2_Click(object sender, EventArgs e)  
    {  
        // Get values from textboxes  
        string productName = textBox1.Text.Trim(); // Product Name  
        string productDescription = richTextBox1.Text.Trim(); // Product Description (multiline support)  
        decimal productPrice;  
        int productStock;  
  
        // Validate inputs  
        if (string.IsNullOrEmpty(productName))  
        {  
            MessageBox.Show("Please enter the product name.");  
            return;  
        }  
  
        if (string.IsNullOrEmpty(productDescription))  
        {  
            MessageBox.Show("Please enter the product description.");  
            return;  
        }  
    }  
}
```

```

if (!decimal.TryParse(textBox2.Text.Trim(), out productPrice) || productPrice <= 0)
{
    MessageBox.Show("Please enter a valid price.");
    return;
}

if (!int.TryParse(textBox3.Text.Trim(), out productStock) || productStock <= 0)
{
    MessageBox.Show("Please enter a valid stock quantity.");
    return;
}

// Connection string
string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
MySQLConnection conn = new MySqlConnection(connstring);

try
{
    conn.Open();

    // Query to insert a new product
    string sql = "INSERT INTO product (Name, Description, Price, Stock) VALUES (@Name,
@Description, @Price, @Stock)";

    MySqlCommand cmd = new MySqlCommand(sql, conn);
    cmd.Parameters.AddWithValue("@Name", productName);
    cmd.Parameters.AddWithValue("@Description", productDescription); // Add
description

```

```
cmd.Parameters.AddWithValue("@Price", productPrice);
cmd.Parameters.AddWithValue("@Stock", productStock);

// Execute the query
int rowsAffected = cmd.ExecuteNonQuery();

if (rowsAffected > 0)
{
    MessageBox.Show("Product added successfully.");

    // Clear the input fields after successful insertion
    textBox1.Clear();
    richTextBox1.Clear(); // Clear description
    textBox2.Clear();
    textBox3.Clear();
}
else
{
    MessageBox.Show("Failed to add the product.");
}
}
catch (Exception ex)
{
    MessageBox.Show("Error: " + ex.Message);
}
finally
{

```

```
        conn.Close();
    }

}

private void textBox1_TextChanged(object sender, EventArgs e)
{

}

private void textBox3_TextChanged(object sender, EventArgs e)
{

}

private void textBox2_TextChanged(object sender, EventArgs e)
{

}

private void button3_Click(object sender, EventArgs e)
{
    this.Hide();
    Form9 newForm = new Form9();

    newForm.Show();
}
```

```

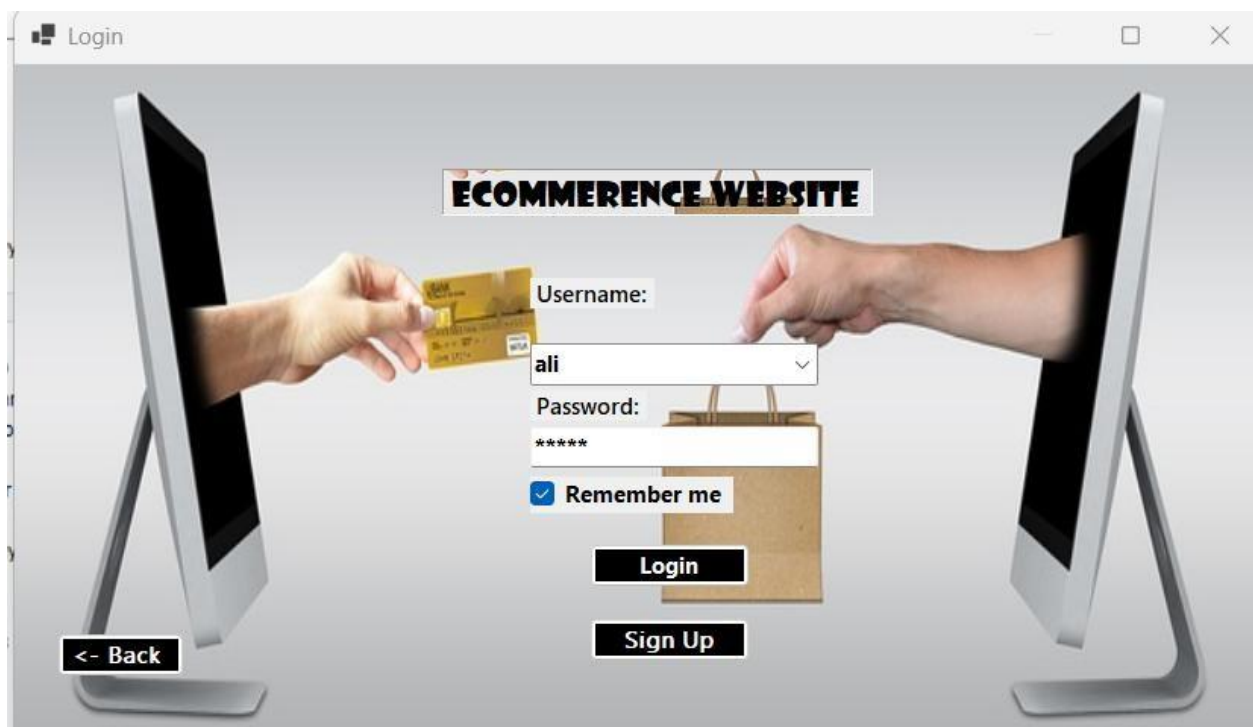
private void Form10_Load(object sender, EventArgs e)
{

}

}
}

```

User ID interface:



Code:

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;

```

```
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using MySql.Data.MySqlClient;
```

```
namespace WinFormsApp2
```

```
{
```

```
    public partial class Form2 : Form
```

```
    {
```

```
        string gender = string.Empty;
```

```
        public Form2()
```

```
        {
```

```
            InitializeComponent();
```

```
        }
```

```
        private void Form2_Load(object sender, EventArgs e)
```

```
        {
```

```
        }
```

```
        private void label3_Click(object sender, EventArgs e)
```

```
        {
```

```
        }
```

```
private void ClearInputs()
{
    textBox1.Clear();
    textBox2.Clear();
    textBox3.Clear();
    textBox4.Clear();
    checkBox1.Checked = false;
    checkBox2.Checked = false;
}

private void button2_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(textBox1.Text))
    {
        MessageBox.Show("Please enter a username.", "Validation Error",
        MessageBoxButtons.OK, MessageBoxIcon.Warning);
        return;
    }

    if (string.IsNullOrEmpty(textBox2.Text))
    {
        MessageBox.Show("Please enter a password.", "Validation Error",
        MessageBoxButtons.OK, MessageBoxIcon.Warning);
        return;
    }

    if (textBox2.Text != textBox4.Text)
```

```
{  
    MessageBox.Show("Passwords do not match.", "Validation Error",  
        MessageBoxButtons.OK, MessageBoxIcon.Warning);  
    return;  
}  
  
if (!checkBox2.Checked && !checkBox1.Checked)  
{  
    MessageBox.Show("Please select a gender.", "Validation Error",  
        MessageBoxButtons.OK, MessageBoxIcon.Warning);  
    return;  
}  
  
if (string.IsNullOrEmpty(textBox3.Text) || !textBox3.Text.Contains("@"))  
{  
    MessageBox.Show("Please enter a valid email address.", "Validation Error",  
        MessageBoxButtons.OK, MessageBoxIcon.Warning);  
    return;  
}  
  
else  
{  
    string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";  
    using (MySqlConnection conn = new MySqlConnection(connstring))  
    {  
        try  
        {  
            conn.Open();  
        }  
    }  
}
```

```

        string sql = "INSERT INTO users (Username, PasswordHash, Gender, Email) VALUES
(@Username, @Password, @Gender, @Email)";

        MySqlCommand cmd = new MySqlCommand(sql, conn);

        // Assign parameters from input fields
        cmd.Parameters.AddWithValue("@Username", textBox1.Text);
        cmd.Parameters.AddWithValue("@Password", textBox2.Text); // No hashing here;
consider adding it for security
        cmd.Parameters.AddWithValue("@Gender", GetSelectedGender());
        cmd.Parameters.AddWithValue("@Email", textBox3.Text);

        cmd.ExecuteNonQuery();

        MessageBox.Show("User registered successfully!", "Success",
MessageBoxButtons.OK, MessageBoxIcon.Information);

        this.Hide();

        Form1 newForm = new Form1();

        // Show Form2
        newForm.Show(); // Non-modal, allows interaction with Form1

        // Optionally clear all inputs
        ClearInputs();
    }
    catch (Exception ex)
    {
        MessageBox.Show($"Error: {ex.Message}", "Database Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
    }

```

```
    }  
    }  
}
```

```
private void button3_Click(object sender, EventArgs e)
```

```
{  
    this.Hide();  
    Form1 newForm = new Form1();
```

```
    // Show Form2  
    newForm.Show();  
}
```

```
private string GetSelectedGender()
```

```
{  
    if (checkBox2.Checked)  
        return "Male";  
    else if (checkBox1.Checked)  
        return "Female";  
    else  
        return "Other";  
}
```

```
private void textBox1_TextChanged(object sender, EventArgs e)
```

```
{  
  
}
```

```
private void textBox2_TextChanged(object sender, EventArgs e)
{
    textBox2.PasswordChar = '*';
}
```

```
private void textBox4_TextChanged(object sender, EventArgs e)
{
    textBox4.PasswordChar = '*';
}
```

```
private void checkBox2_CheckedChanged(object sender, EventArgs e)
{
```

```
    if (checkBox2.Checked)
```

```
    {
```

```
        // If the Female checkbox is checked, set the gender to Female
```

```
        gender = "Male";
```

```
        // Optional: You can uncheck the Male checkbox to ensure only one gender is selected
        at a time
```

```
        checkBox1.Checked = false;
```

```
    }
```

```
}
```

```
private void checkBox1_CheckedChanged(object sender, EventArgs e)
```

```
{
```

```
    if (checkBox2.Checked)
```

```
    {
```

```
        // If the Female checkbox is checked, set the gender to Female
```

```
gender = "Female";
```

```
// Optional: You can uncheck the Male checkbox to ensure only one gender is selected  
at a time
```

```
checkBox1.Checked = false;
```

```
}
```

```
}
```

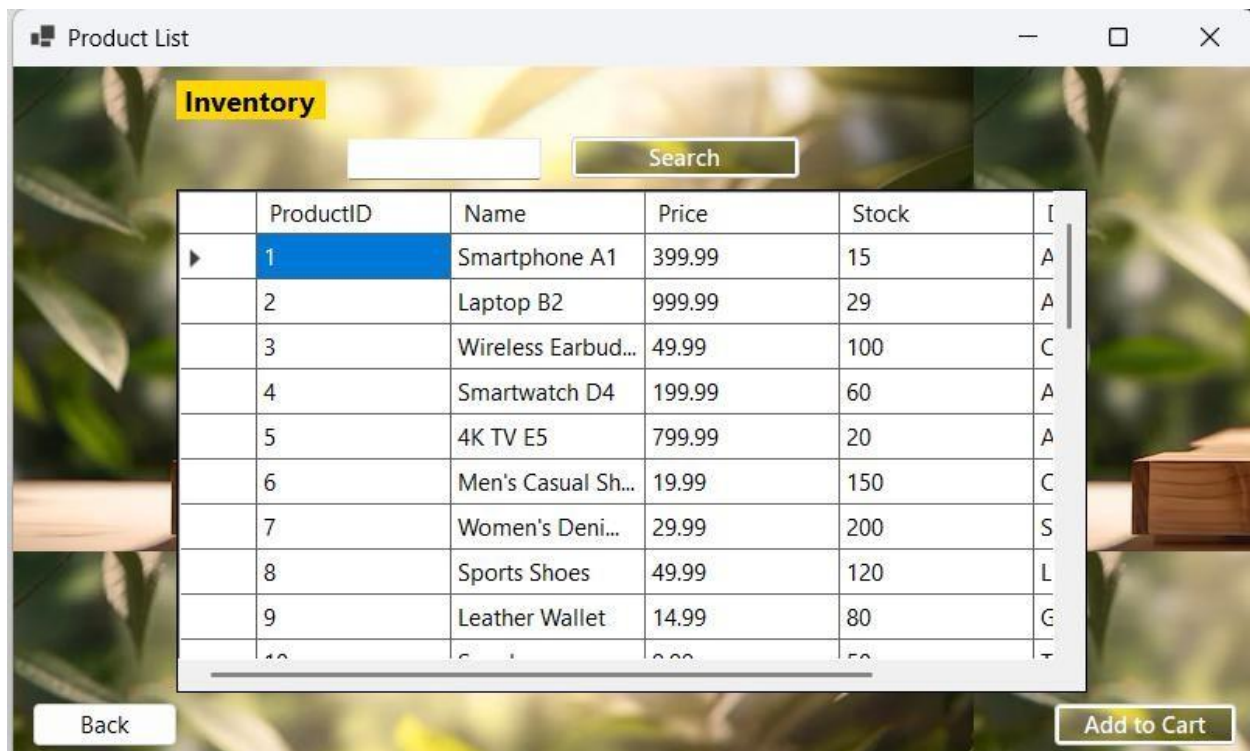
```
private void textBox3_TextChanged(object sender, EventArgs e)
```

```
{
```

```
}
```

```
}
```

```
}
```



Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using MySql.Data.MySqlClient;

namespace WinFormsApp2
{
```

```
public partial class Form3 : Form
{
    public Form3()
    {
        InitializeComponent();
    }

    private void Form3_Load(object sender, EventArgs e)
    {
        LoadProductData();
    }

    private DataTable products = null; // Declare a global DataTable

    private void LoadProductData()
    {
        string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
        MySqlConnection conn = new MySqlConnection(connstring);

        try
        {
            conn.Open(); // Open the connection

            // Query to fetch product data
            string sql = "SELECT * FROM product";

            // Create a data adapter
            MySqlDataAdapter dataAdapter = new MySqlDataAdapter(sql, conn);
```

```
// Initialize the global DataTable
products = new DataTable();

// Fill the DataTable with data from the database
dataAdapter.Fill(products);

// Bind the DataTable to the DataGridView
dataGridView1.DataSource = products;
}
catch (Exception ex)
{
    // Show any errors that occur
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    conn.Close(); // Close the connection
}
}

private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
{
}

private void button1_Click(object sender, EventArgs e)
{
}
```

```

this.Hide();

Form4 newForm = new Form4();

// Show Form2
newForm.Show();
}

private void textBox1_TextChanged_1(object sender, EventArgs e)
{
    string filterText = textBox1.Text.Trim();

    // Escape single quotes for the filter
    filterText = filterText.Replace("'", "");

    if (string.IsNullOrEmpty(filterText))
    {
        // If the search box is empty, show the original data
        dataGridView1.DataSource = products;
    }
    else
    {
        // Apply a filter to the global DataTable
        DataView filteredView = new DataView(products);

        // Construct a filter for all columns
        filteredView.RowFilter = $"@\"
        CONVERT(ProductID, 'System.String') LIKE '%{filterText}%' OR

```

```
        Name LIKE '%{filterText}%' OR
        CONVERT(Price, 'System.String') LIKE '%{filterText}%' OR
        CONVERT(Stock, 'System.String') LIKE '%{filterText}%'
";

        dataGridView1.DataSource = filteredView;
    }
}

private void button2_Click(object sender, EventArgs e)
{

}

private void button3_Click(object sender, EventArgs e)
{
    Form7 form = new Form7();
    form.Show();
    this.Hide();
}
}
```

ProductID	Name	Price	Stock
1	Smartphone A1	399.99	15
2	Laptop B2	999.99	29
3	Wireless Earbud...	49.99	100
4	Smartwatch D4	199.99	60
5	4K TV E5	799.99	20
6	Men's Casual Sh...	19.99	150
7	Women's Deni...	29.99	200
8	Sports Shoes	49.99	120
9	Leather Wallet	14.99	80

Enter Id: Quantity:

<- Back Add to Cart Continue ->

Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using MySql.Data.MySqlClient;

namespace WinFormsApp2
{
    public partial class Form4 : Form
```

```
{  
    public Form4()  
    {  
        InitializeComponent();  
        this.Load += new System.EventHandler(this.Form4_Load);  
    }  
  
    private void button3_Click(object sender, EventArgs e)  
    {  
        this.Hide();  
        Form3 newForm = new Form3();  
  
        // Show Form2  
        newForm.Show();  
    }  
    private void Form4_Load(object sender, EventArgs e)  
    {  
        LoadProductData();  
    }  
    private void LoadProductData()  
    {  
        string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";  
        MySqlConnection conn = new MySqlConnection(connstring);  
  
        try  
        {
```

```
conn.Open(); // Open the connection

// Query to fetch product data
string sql = "SELECT * FROM product";

// Create a data adapter
MySqlDataAdapter dataAdapter = new MySqlDataAdapter(sql, conn);

// Create a DataTable to hold the data
DataTable dataTable = new DataTable();

// Fill the DataTable with data from the database
dataAdapter.Fill(dataTable);

// Bind the DataTable to the DataGridView
dataGridView1.DataSource = dataTable;
}
catch (Exception ex)
{
    // Show any errors that occur
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    conn.Close(); // Close the connection
}
}
```

```
private void button1_Click(object sender, EventArgs e)
{
    this.Hide();

    Form6 newForm = new Form6();

    // Show Form2
    newForm.Show();
}

private void dataGridView1_CellContentClick_1(object sender, DataGridViewCellEventArgs
e)
{
}

private void Form4_Load_1(object sender, EventArgs e)
{
}

private void button2_Click(object sender, EventArgs e)
{
    // Get ProductID and Quantity from the textboxes
    int productId = Convert.ToInt32(textBox1.Text);
    int quantity = Convert.ToInt32(textBox2.Text);

    // Define your price limit (optional, you can adjust this as needed)
```

```
decimal priceLimit = 10000.00m;

// Fetch the price of the product based on ProductID
decimal productPrice = GetProductPrice(productId);

if (productPrice == -1)
{
    // If the price is not found, show an error message
    MessageBox.Show("Product not found.");
    return;
}

if (quantity <= 0)
{
    // If the price is not found, show an error message
    MessageBox.Show("Please Enter some quantity to be added to stock.");
    return;
}

// Calculate the total price for the given quantity
decimal totalPrice = productPrice * quantity;

// Check if the total price exceeds the limit
if (totalPrice > priceLimit)
{
    MessageBox.Show("The total price exceeds the allowed limit.");
    return;
}
```

```

// Proceed to insert into the Cart table
InsertIntoCart(productId, quantity, productPrice, totalPrice);

textBox1.Clear(); // Clears the content of textBox1
textBox2.Clear(); // Clears the content of textBox2
}

private decimal GetProductPrice(int productId)
{
    // Connection string to your MySQL database
    string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
    MySqlConnection conn = new MySqlConnection(connstring);
    decimal price = -1; // Default value if the product is not found

    try
    {
        conn.Open();

        // Query to fetch the price of the product based on ProductID
        string sql = "SELECT Price FROM product WHERE ProductID = @ProductID";
        MySqlCommand cmd = new MySqlCommand(sql, conn);
        cmd.Parameters.AddWithValue("@ProductID", productId);

        // Execute the query and get the result
        object result = cmd.ExecuteScalar();

        if (result != DBNull.Value && result != null)
        {

```

```

        price = Convert.ToDecimal(result);
    }
}
catch (Exception ex)
{
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    conn.Close();
}

return price;
}

private void InsertIntoCart(int productId, int quantity, decimal productPrice, decimal
totalPrice)
{
    int userId = 1; // Example: Assuming user ID is 1, replace it with actual logic

    // Get the values from the text boxes
    int enteredProductId = Convert.ToInt32(textBox1.Text); // ProductID from the input
textbox
    int enteredQuantity = Convert.ToInt32(textBox2.Text); // Quantity from the input textbox

    // Query to get the product price from the database
    decimal enteredProductPrice = 0.00m;
    string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
    MySqlConnection conn = new MySqlConnection(connstring);

```

```
try
{
    conn.Open();

    // Query to get the product price from the product table using ProductID
    string priceQuery = "SELECT Price FROM product WHERE ProductID = @ProductID";
    MySqlCommand priceCmd = new MySqlCommand(priceQuery, conn);
    priceCmd.Parameters.AddWithValue("@ProductID", enteredProductId);

    // Execute the query and retrieve the price
    object result = priceCmd.ExecuteScalar();
    if (result != null)
    {
        enteredProductPrice = Convert.ToDecimal(result);
    }
    else
    {
        MessageBox.Show("Product not found.");
        return;
    }

    // Now check if the quantity does not exceed the available stock
    string stockQuery = "SELECT Stock FROM product WHERE ProductID = @ProductID";
    MySqlCommand stockCmd = new MySqlCommand(stockQuery, conn);
    stockCmd.Parameters.AddWithValue("@ProductID", enteredProductId);
```

```
// Execute the query and retrieve the stock quantity
object stockResult = stockCmd.ExecuteScalar();
if (stockResult != null)
{
    int availableStock = Convert.ToInt32(stockResult);
    if (enteredQuantity > availableStock)
    {
        MessageBox.Show("Insufficient stock available.");
        return;
    }
}
```

```
// Query to insert data into the Cart table (no need for TotalPrice in the INSERT statement)
```

```
string insertQuery = "INSERT INTO Cart (UserID, ProductID, Quantity, Price) " +
    "VALUES (@UserID, @ProductID, @Quantity, @Price)";
```

```
MySqlCommand cmd = new MySqlCommand(insertQuery, conn);
cmd.Parameters.AddWithValue("@UserID", userId);
cmd.Parameters.AddWithValue("@ProductID", enteredProductId);
cmd.Parameters.AddWithValue("@Quantity", enteredQuantity);
cmd.Parameters.AddWithValue("@Price", enteredProductPrice);
```

```
// Execute the query to insert the data into the Cart table
cmd.ExecuteNonQuery();
```

```
// Now reduce the stock in the product table
```

```

        string updateStockQuery = "UPDATE product SET Stock = Stock - @Quantity WHERE
ProductID = @ProductID";

        MySqlCommand updateStockCmd = new MySqlCommand(updateStockQuery, conn);
        updateStockCmd.Parameters.AddWithValue("@Quantity", enteredQuantity);
        updateStockCmd.Parameters.AddWithValue("@ProductID", enteredProductId);

        // Execute the query to update the stock
        updateStockCmd.ExecuteNonQuery();
        LoadProductData();
        // Show a message confirming the insertion
        MessageBox.Show("Product added to cart and stock updated successfully.");
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
    finally
    {
        conn.Close();
    }
}

private void textBox1_TextChanged(object sender, EventArgs e)
{
}

```

```

private void textBox2_TextChanged(object sender, EventArgs e)
{

}

}
}

```

Shopping Cart Form

Cart Table

	CartID	UserID	ProductID
▶	9	1	1
•			

Remove Items

Enter Id.

Remove

<- Back

Total Amount: \$3,999.90

Continue ->

Code:

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;

```

```
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using MySql.Data.MySqlClient;
```

```
namespace WinFormsApp2
```

```
{
```

```
    public partial class Form6 : Form
```

```
    {
```

```
        public Form6()
```

```
        {
```

```
            InitializeComponent();
```

```
        }
```

```
        private void Form6_Load(object sender, EventArgs e)
```

```
        {
```

```
            LoadProductData();
```

```
            CalculateTotalAmount();
```

```
        }
```

```
        private void LoadProductData()
```

```
        {
```

```
            string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
```

```
            MySqlConnection conn = new MySqlConnection(connstring);
```

```
            try
```

```
{  
    conn.Open(); // Open the connection  
  
    // Query to fetch product data  
    string sql = "SELECT * FROM cart";  
  
    // Create a data adapter  
    MySqlDataAdapter dataAdapter = new MySqlDataAdapter(sql, conn);  
  
    // Create a DataTable to hold the data  
    DataTable dataTable = new DataTable();  
  
    // Fill the DataTable with data from the database  
    dataAdapter.Fill(dataTable);  
  
    // Bind the DataTable to the DataGridView  
    dataGridView1.DataSource = dataTable;  
}  
catch (Exception ex)  
{  
    // Show any errors that occur  
    MessageBox.Show("Error: " + ex.Message);  
}  
finally  
{  
    conn.Close(); // Close the connection  
}
```

```
}
```

```
private void button3_Click(object sender, EventArgs e)
```

```
{
```

```
    this.Hide();
```

```
    Form4 newForm = new Form4();
```

```
    // Show Form2
```

```
    newForm.Show();
```

```
}
```

```
private void button1_Click(object sender, EventArgs e)
```

```
{
```

```
    this.Hide();
```

```
    Form5 newForm = new Form5();
```

```
    // Show Form2
```

```
    newForm.Show();
```

```
}
```

```
private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
```

```
{
```

```
}
```

```
private void CalculateTotalAmount()
```

```
{
```

```
    // Connection string to your MySQL database
```

```
string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
MySQLConnection conn = new MySqlConnection(connstring);

try
{
    // Open the connection
    conn.Open();

    // SQL query to get the total amount from the Cart table
    string sql = "SELECT SUM(TotalPrice) FROM Cart";

    // Create a MySQL command to execute the query
    MySqlCommand cmd = new MySqlCommand(sql, conn);

    // Execute the query and get the result
    object result = cmd.ExecuteScalar();

    // Check if the result is not null (to avoid errors if no data exists)
    if (result != DBNull.Value)
    {
        // Convert the result to a decimal and assign it to the textBox2
        decimal totalAmount = Convert.ToDecimal(result);
        textBox2.Text = totalAmount.ToString("C"); // Format as currency (optional)
    }
    else
    {
        // In case no total price is found, set textBox2 to 0
    }
}
```

```
        textBox2.Text = "0.00";
    }
}
catch (Exception ex)
{
    // Handle any errors that occur during the process
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    // Close the connection to the database
    conn.Close();
}
}
private void textBox2_TextChanged(object sender, EventArgs e)
{
}

private void textBox1_TextChanged(object sender, EventArgs e)
{
}

private void label5_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(textBox1.Text))
    {
```

```
        return; // Do nothing if the text box is empty
    }

    int enteredProductId = Convert.ToInt32(textBox1.Text); // Get the ProductID from
textBox1

    string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
    MySqlConnection conn = new MySqlConnection(connstring);

    try
    {
        conn.Open();

        // Query to check if the ProductID exists in the Cart table
        string cartQuery = "SELECT Quantity FROM Cart WHERE ProductID = @ProductID";
        MySqlCommand cartCmd = new MySqlCommand(cartQuery, conn);
        cartCmd.Parameters.AddWithValue("@ProductID", enteredProductId);

        // Execute the query and retrieve the cart quantity
        object cartResult = cartCmd.ExecuteScalar();

        if (cartResult != null)
        {
            int cartQuantity = Convert.ToInt32(cartResult);

            // Query to remove the product from the Cart table
            string deleteCartQuery = "DELETE FROM Cart WHERE ProductID = @ProductID";
            MySqlCommand deleteCartCmd = new MySqlCommand(deleteCartQuery, conn);
```

```

deleteCartCmd.Parameters.AddWithValue("@ProductID", enteredProductId);

// Execute the delete query to remove the product from the cart
deleteCartCmd.ExecuteNonQuery();

// Query to update the stock in the Products table
string updateStockQuery = "UPDATE product SET Stock = Stock + @Quantity WHERE
ProductID = @ProductID";

SqlCommand updateStockCmd = new SqlCommand(updateStockQuery, conn);

updateStockCmd.Parameters.AddWithValue("@Quantity", cartQuantity); // Add the
cart quantity back to the stock

updateStockCmd.Parameters.AddWithValue("@ProductID", enteredProductId);

// Execute the query to update the stock
updateStockCmd.ExecuteNonQuery();

// Refresh the DataGridView or any UI element to show updated stock
LoadProductData();
CalculateTotalAmount();

MessageBox.Show("Product removed from cart and stock updated.");
textBox1.Clear();
}
else
{
    MessageBox.Show("Product not found in the cart.");
}
}

```

```

catch (Exception ex)
{
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    conn.Close();
}

}
}
}

```

Order Placement Form

OrderList

	CartID	UserID	ProductID
	9	1	1
*			

Back to Cart

Payment

☐ Cash

Confirm Order

Total Amount:

\$3,999.90

Code:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using MySql.Data.MySqlClient;

namespace WinFormsApp2
{
    public partial class Form5 : Form
    {
        public Form5()
        {
            InitializeComponent();
            this.Load += new System.EventHandler(this.Form5_Load);
        }

        private void button2_Click(object sender, EventArgs e)
        {
            this.Hide();
            Form6 newForm = new Form6();
```

```
// Show Form2
newForm.Show(); // Non-modal, allows interaction with Form1
}

private void Form5_Load(object sender, EventArgs e)
{
    LoadProductData();
    CalculateTotalAmount();
}

private void LoadProductData()
{
    string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
    MySqlConnection conn = new MySqlConnection(connstring);

    try
    {
        conn.Open(); // Open the connection

        // Query to fetch product data
        string sql = "SELECT * FROM cart";

        // Create a data adapter
        MySqlDataAdapter dataAdapter = new MySqlDataAdapter(sql, conn);

        // Create a DataTable to hold the data
        DataTable dataTable = new DataTable();

        // Fill the DataTable with data from the database
```

```
        dataAdapter.Fill(dataTable);

        // Bind the DataTable to the DataGridView
        dataGridView1.DataSource = dataTable;
    }
    catch (Exception ex)
    {
        // Show any errors that occur
        MessageBox.Show("Error: " + ex.Message);
    }
    finally
    {
        conn.Close(); // Close the connection
    }
}

private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
{
}

private void Form5_Load_1(object sender, EventArgs e)
{
}

private void CalculateTotalAmount()
{
    // Connection string to your MySQL database
```

```
string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";
MySQLConnection conn = new MySqlConnection(connstring);

try
{
    // Open the connection
    conn.Open();

    // SQL query to get the total amount from the Cart table
    string sql = "SELECT SUM(TotalPrice) FROM Cart";

    // Create a MySQL command to execute the query
    MySqlCommand cmd = new MySqlCommand(sql, conn);

    // Execute the query and get the result
    object result = cmd.ExecuteScalar();

    // Check if the result is not null (to avoid errors if no data exists)
    if (result != DBNull.Value)
    {
        // Convert the result to a decimal and assign it to the textBox2
        decimal totalAmount = Convert.ToDecimal(result);
        textBox2.Text = totalAmount.ToString("C"); // Format as currency (optional)
    }
    else
    {
        // In case no total price is found, set textBox2 to 0
    }
}
```

```
        textBox2.Text = "0.00";
    }
}
catch (Exception ex)
{
    // Handle any errors that occur during the process
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    // Close the connection to the database
    conn.Close();
}
}
private void textBox2_TextChanged(object sender, EventArgs e)
{
}

private void label2_Click(object sender, EventArgs e)
{
    // Ensure the checkbox is selected
    if (!checkBox1.Checked)
    {
        MessageBox.Show("Please select payment method.");
        return;
    }
}
```

```

// Connection string
string connstring = "server=localhost;uid=root;pwd=root;database=emp_cust";

 MySqlConnection conn = new MySqlConnection(connstring);

try
{
    conn.Open();

    // Calculate the total amount for the current user's cart
    string totalAmountQuery = "SELECT SUM(Quantity * Price) AS TotalAmount FROM cart
WHERE UserID = @UserID";

    MySqlCommand totalAmountCmd = new MySqlCommand(totalAmountQuery, conn);
    totalAmountCmd.Parameters.AddWithValue("@UserID", 1); // Replace '1' with the
actual user ID logic

    object totalAmountResult = totalAmountCmd.ExecuteScalar();

    decimal totalAmount = totalAmountResult != null && totalAmountResult !=
DBNull.Value
        ? Convert.ToDecimal(totalAmountResult)
        : 0;

    // Check if total amount is less than or equal to zero
    if (totalAmount <= 0)
    {
        MessageBox.Show("Total amount is zero or invalid. Cannot confirm the order.");
        return;
    }
}

```

```

    }

    // Confirm the order: Clear the cart (delete rows) after successful order placement
    string clearCartQuery = "DELETE FROM cart WHERE UserID = @UserID";
    MySqlCommand clearCartCmd = new MySqlCommand(clearCartQuery, conn);
    clearCartCmd.Parameters.AddWithValue("@UserID", 1); // Replace '1' with the actual
user ID logic

    clearCartCmd.ExecuteNonQuery();

    // Notify the user about the successful order confirmation
    MessageBox.Show($"Order confirmed successfully! Total Amount: {totalAmount:C}");
    LoadProductData();
    CalculateTotalAmount();
}
catch (Exception ex)
{
    MessageBox.Show("Error: " + ex.Message);
}
finally
{
    conn.Close();
}
}

```

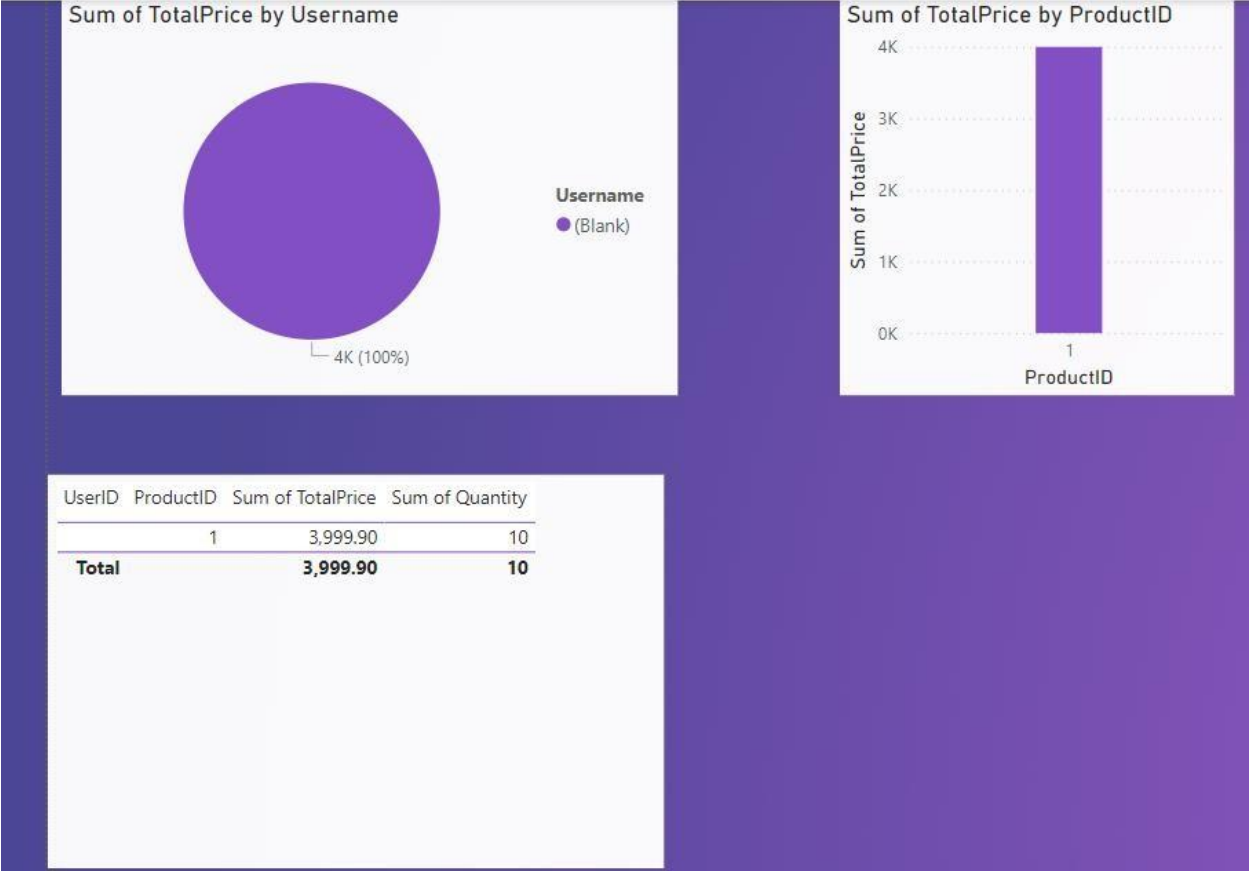
```

private void checkBox1_CheckedChanged(object sender, EventArgs e)

```

```
{  
  }  
}  
}
```

Power BI



Modelling:

